



I/O Multiplexing



Multiplexing is combining multiple signals into one medium, then separating them at the other end.

The word comes from Latin:

Multi (many) + Plex (fold) = "folding many into one"

Example of Frequency-Division Multiplexing (FDM) in telecommunications:

```
Radio Station 1: ~~~~~ (91.5 MHz)
Radio Station 2: ~~~~~ (93.7 MHz)
Radio Station 3: ~~~~~ (96.1 MHz)
      ||||| (combined into one signal through the air)
Your radio: Separates them by tuning to specific frequency
```

Same principle applies to I/O Multiplexing:

```
Client 1: "GET /index.html"
Client 2: "POST /submit"
Client 3: "GET /image.png"
      ||||| (combined into ONE poll() system call)
Your server: Separates them by checking which FD is ready
```

I/O Multiplexing allows a single thread to monitor multiple file descriptors simultaneously, waiting until at least one is ready for reading or writing.

Comparison

Without multiplexing:

```
while (true) {
    int clientFd = accept(listenFd, ...); // Accept one client
    char buf[4096];
    recv(clientFd, buf, ...); // Wait for THIS client (blocks)
    send(clientFd, response, ...); // Send response
    close(clientFd); // One client at a time!
}
```

With multiplexing:

```
while (true) {
    poll(pollFds, count, -1); // Wait for ANY client
    for (each ready fd) {
        if (fd is listenFd) accept();
        if (fd has data) recv();
        if (fd can write) send();
    }
}
```